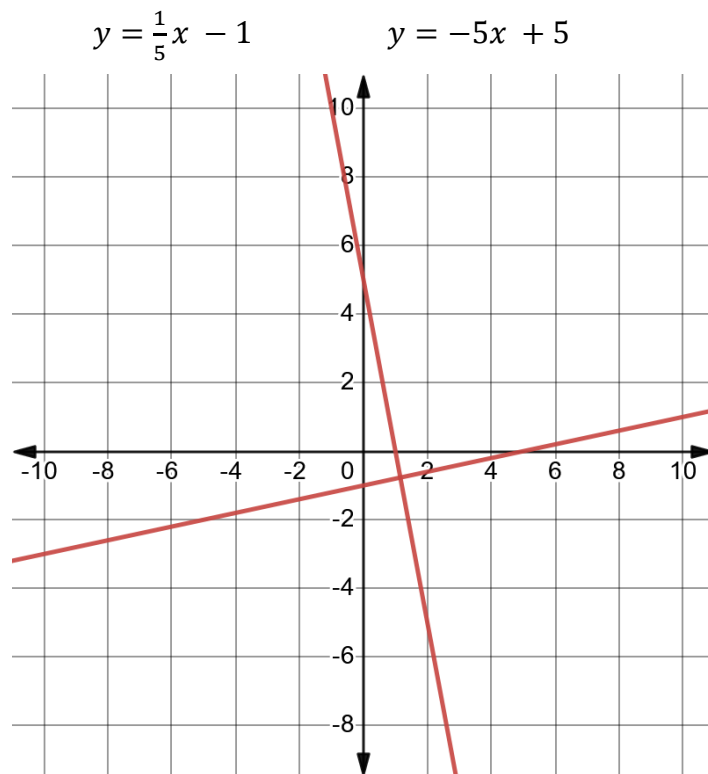


Geometry
Unit 13
Lesson 1 Homework

Name Answer Key
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1. Graph the equations on the coordinate plane.



2. The equations $y = \frac{1}{5}x - 1$ and $y = -5x + 5$ resulted in _____ lines.

☐ parallel
☒ perpendicular
☐ vertical

3. The slopes of the equations are _____

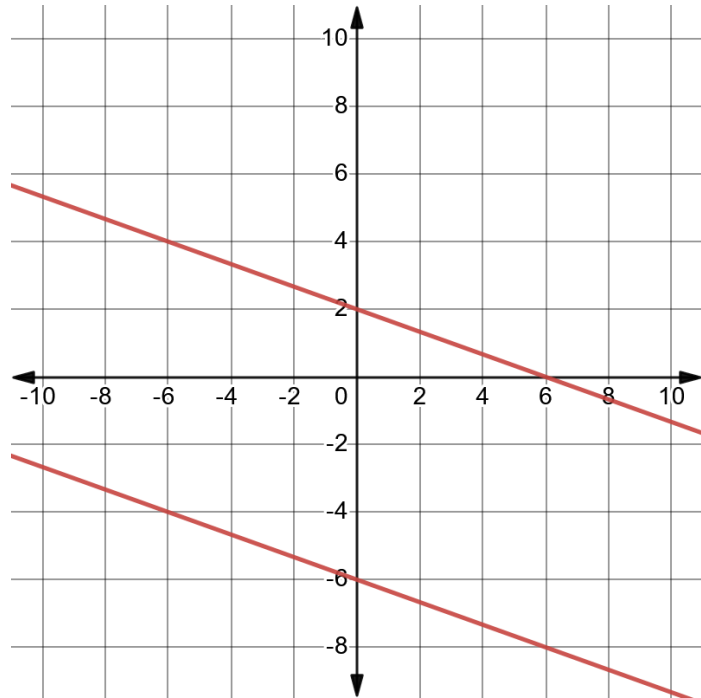
☐ equal.
☐ undefined.
☒ opposite reciprocals.

4. Rewrite the equation $x + 3y = -18$ in slope intercept form and graph the equation using slope-intercept form.

$$x + 3y = -18$$

$$3y = -x - 18$$

$$y = -\frac{1}{3}x - 6$$



5. Determine if $x + 3y = -18$ is parallel, perpendicular, or neither to the graph of the equation $y = -\frac{1}{3}x + 2$.

Parallel. The slopes are the same.

6. Graph the equation $y = -\frac{1}{3}x + 2$ on the graph to verify your answer to 5.

See graph.

7. Which equation represents a line which is perpendicular to the line $y = -\frac{8}{3}x - 1$?

A. $8x + 3y = 15$

B. $8y - 3x = 48$

C. $-3x - 8y = -56$

D. $8x - 3y = -24$

$$8y - 3x = 48$$

$$8y = 3x + 48$$

$$y = \frac{3}{8}x + 6$$

8. Which equation represents a line which is parallel to the line $3x + y = -40$?

A. $y = -3x + 3$

B. $y = 3x - 5$

C. $y = -\frac{1}{3}x + 8$

D. $y = \frac{1}{3}x - 2$

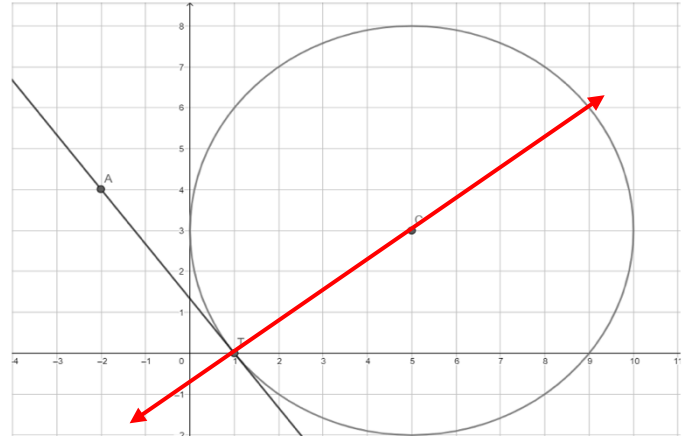
$$3x + y = -40$$

$$y = -3x - 40$$

Geometry
Unit 13
Lesson 2 Homework

Name Answer Key
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Use the following information for questions 1-4. Circle C is shown, where point T is on the circle and line AT is tangent to Circle C .



1. Use a straightedge to draw a line that goes through points T and C .

2. Write the equation of line AT .

$$y = -\frac{4}{3}x + \frac{4}{3}$$

3. Write the equation of line segment CT .

$$y = \frac{3}{4}x - \frac{3}{4}$$

4. Line AT and line CT intersect at point $(1, 0)$. Use the equations of line AT and CT to confirm this point.

$$\begin{aligned} y &= -\frac{4}{3}x + \frac{4}{3} \\ 0 &= -\frac{4}{3}(1) + \frac{4}{3} \\ 0 &= 0 \end{aligned}$$

$$\begin{aligned} y &= \frac{3}{4}x - \frac{3}{4} \\ 0 &= \frac{3}{4}(1) - \frac{3}{4} \\ 0 &= 0 \end{aligned}$$

Use the following information for questions 5-8. Circle N has a center at $(2, 3)$ and a tangent line with the equation $y = \frac{1}{2}x + 7$.

5. Write the equation of the line perpendicular to the tangent line.

$y = -2x + 7$

6. Find the point where the tangent line touches Circle N . Call this point D .

$\frac{1}{2}x + 7 = -2x + 7$ $y = -2(0) + 7$
 $x = 0$ $y = 7$
 $(0, 7)$

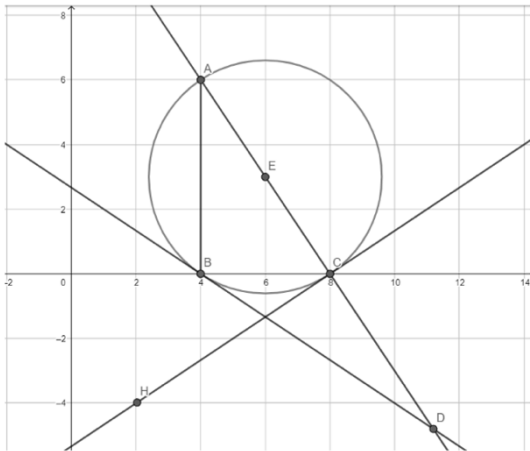
7. What is the radius of Circle N ?

$r = \sqrt{(2 - 0)^2 + (3 - 7)^2} = \sqrt{20} = 2\sqrt{5}$

8. Write the equation of Circle N .

$(x - 2)^2 + (y - 3)^2 = 20$

Use the following information for questions 9-10. Circle E is shown with secant line AD , tangent lines DB and CH , and chord AB .



9. Complete the table.

Line Segment	Length
$\overline{AB}$	6
$\overline{BC}$	4
$\overline{AC}$	$2\sqrt{13}$

10. Write the equation for circle E .

$(x - 6)^2 + (y - 3)^2 = 13$

Geometry
Unit 13
Lesson 3 Homework

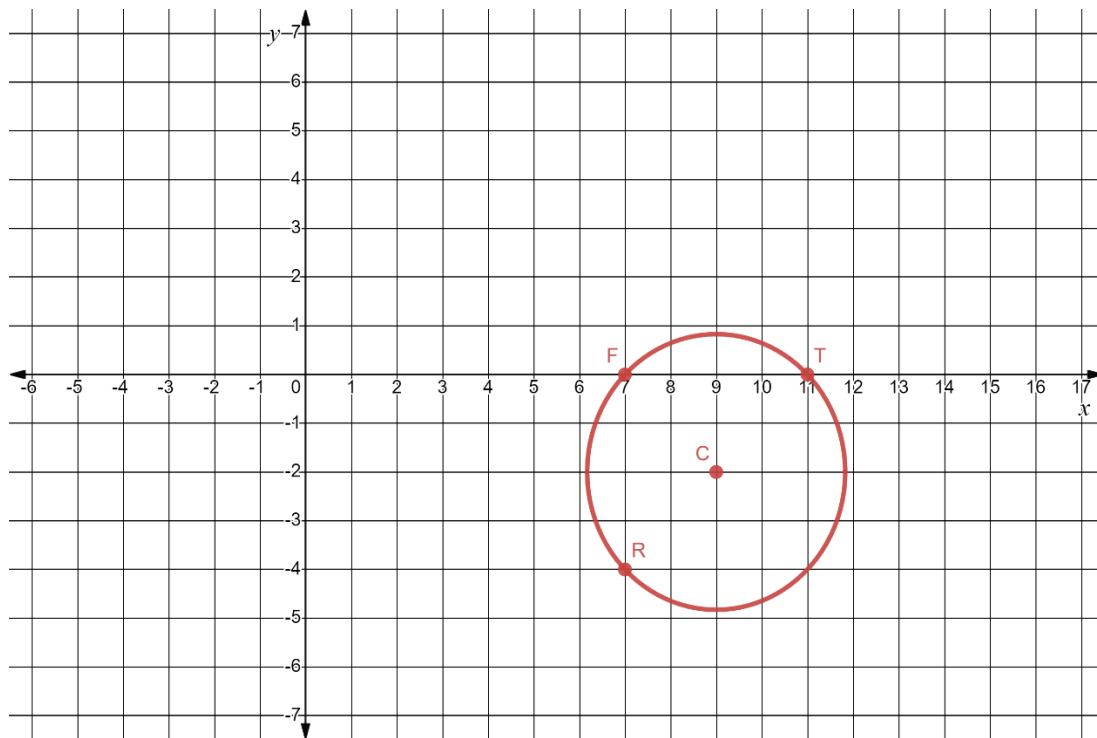
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Use the following information for questions 1-4. The equation of Circle C is $(x - 9)^2 + (y + 2)^2 = 8$.

1. Complete the table by substituting the given coordinates for x and y in the equation of Circle C . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$F(7,0)$	$(7 - 9)^2 + (0 + 2)^2 = (-2)^2 + (2)^2$ $= 4 + 4 = 8$	$8 = 8$
$R(7, -4)$	$(7 - 9)^2 + (-4 + 2)^2 = (-2)^2 + (-2)^2$ $= 4 + 4 = 8$	$8 = 8$
$T(11,0)$	$(11 - 9)^2 + (0 + 2)^2 = (2)^2 + (2)^2$ $= 4 + 4 = 8$	$8 = 8$

2. Graph Circle C , $(x - 9)^2 + (y + 2)^2 = 8$.



3. Plot $F(7,0)$, $R(7,-4)$, $T(11,0)$ on the circle.
4. Determine the length of the diameter of Circle C .

$$d = 2 \cdot \sqrt{8} = 4\sqrt{2} \approx 5.66$$

Use the following information for questions 5-7. Circle P has a center located at $T(3,4)$ and point $L(4,9)$ on the circle.

5. Determine two points that are the endpoints of a diameter of Circle P .

Answers may vary. $(3,4) = \left(\frac{4+x}{2}, \frac{9+y}{2}\right) \rightarrow \begin{array}{l} 4+x=6 \\ x=2 \end{array} \quad \begin{array}{l} 9+y=8 \\ y=-1 \end{array}$
 $(4,9)$ and $(2,-1)$

6. What is the length of the radius?

$T(3,4)$ and $L(4,9)$
 $\sqrt{(3-4)^2 + (4-9)^2} = \sqrt{26}$

7. Write the equation for Circle P .

$(x-3)^2 + (y-4)^2 = 26$

The equation of Circle W is $(x-3)^2 + (y+5)^2 = 17$. Determine whether the following points lie inside, outside, or on Circle W .

Point	Work	Location
$C(3, -10)$	$(3-3)^2 + (-10+5)^2 = 25 > 17$	☐ Inside Circle W ☒ Outside Circle W ☐ On Circle W
$N(2,3)$	$(2-3)^2 + (3+5)^2 = 65 > 17$	☐ Inside Circle W ☒ Outside Circle W ☐ On Circle W
$T(4,-1)$	$(4-3)^2 + (-1+5)^2 = 17 = 17$	☐ Inside Circle W ☐ Outside Circle W ☒ On Circle W

8. The equation of Circle R is $(x+3)^2 + (y+6)^2 = 10$.

Select all points that are outside Circle R .

- ☐ $(-3, -6)$ $(-3+3)^2 + (-6+6)^2 = 0 < 10$ inside
☒ $(-3, -2)$ $(-3+3)^2 + (-2+6)^2 = 16 > 10$ outside
☐ $(0, -5)$ $(0+3)^2 + (-5+6)^2 = 10 = 10$ on
☒ $(0, 0)$ $(0+3)^2 + (0+6)^2 = 45 > 10$ outside
☐ $(-1, -6)$ $(-1+3)^2 + (-6+6)^2 = 4 < 10$ inside

9. Circle M and Circle T intersect at point H .

Equation of Circle M is $x^2 + (y-2)^2 = 1$

Equation of Circle T is $(x+3)^2 + (y-1)^2 = 9$

Which coordinate pair is the location of point H ?

- ☐ $(-3, 4)$ $(-3)^2 + (4-2)^2 = 13 > 1 \rightarrow$ not on circle M
☐ $(4, 0)$ $(4)^2 + (0-2)^2 = 20 > 1 \rightarrow$ not on circle M
☐ $(0, 3)$ $(0)^2 + (3-2)^2 = 1 \checkmark$, $(0+3)^2 + (3-1)^2 = 13 > 9 \rightarrow$ not on circle T
☒ $(0, 1)$ $(0)^2 + (1-2)^2 = 1 = 1 \checkmark$, $(0+3)^2 + (1-1)^2 = 9 = 9 \checkmark$

Use the following information for questions 1-4. Directed line segment $\overline{WZ}$.



- 1. How many equal parts is $\overline{WZ}$ partitioned into?
11
- 2. Make a mark where $\overline{WZ}$ is partitioned into a 2:3 ratio. Call this point X .
 $2 + 3 = 5$
 $2 \left(\frac{11}{5} \right) = 4 \frac{2}{5}$
- 3. Complete the statement.

Point X is $\frac{2}{5}$ of the distance from W to Z .

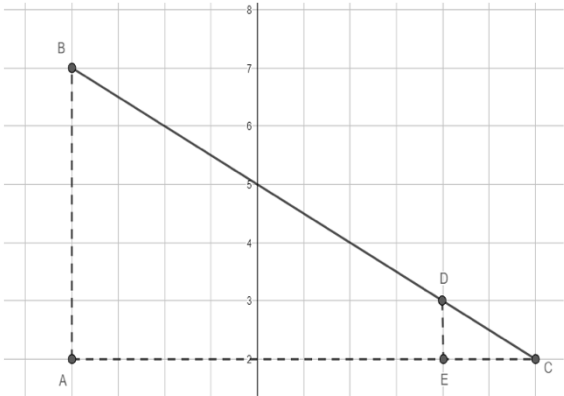
- 4. Complete the equation that can be used to find the distance of $\overline{WX}$.

$WX = \frac{2}{5} WZ$

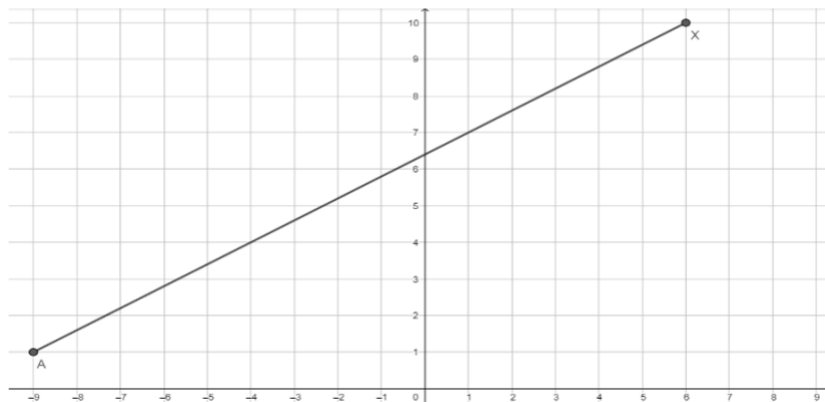
Point D partitions directed line segment $\overline{BC}$. Points $A(-4,2)$, $E(4,2)$, $D(4,3)$, $B(-4,7)$, and $C(6,2)$ are shown on the coordinate grid.

Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represents.

Segment Ratio	Numeric Ratio	Type of Ratio
$CD:DB$	1:4	☒ part: part ☐ part: whole
$CE:CA$	1:5	☐ part: part ☒ part: whole
$DE:BA$	1:5	☐ part: part ☒ part: whole
$CD:CB$	1:5	☐ part: part ☒ part: whole



Use the following information for questions 5-7. Directed line segment AX is shown with endpoints at point $A(-9,1)$ and point $X(6,10)$.



Directed line segment AX is partitioned $\frac{4}{5}$ of the distance from A to X at point B .

5. What is the horizontal distance of $\overline{AB}$?

$$\frac{4}{5}(15) = 12$$

6. What is the vertical distance of $\overline{AB}$?

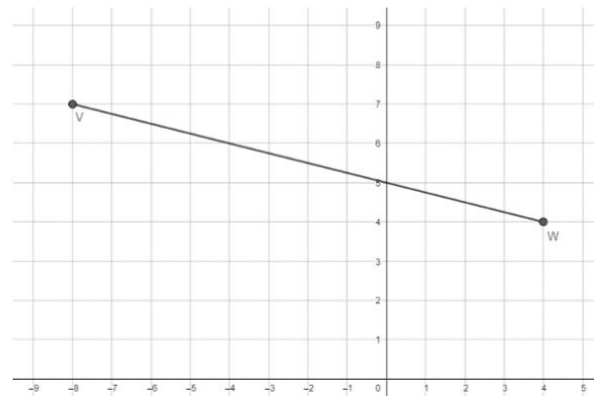
$$\frac{4}{5}(9) = \frac{36}{5} \text{ or } 7.2$$

7. What are the coordinates of point B ?

$$(-9 + 12, 1 + 7.2) = (3, 8.2)$$

Use the following information for questions 8-10.

Directed line segment VW has endpoints at $V(-8,7)$ and $W(4,4)$.



Directed line segment $\overline{VW}$ is partitioned $\frac{1}{4}$ of the distance from V to W at point R .

8. What is the horizontal distance of $\overline{VR}$?

$$\frac{1}{4}(4 - (-8)) = 3$$

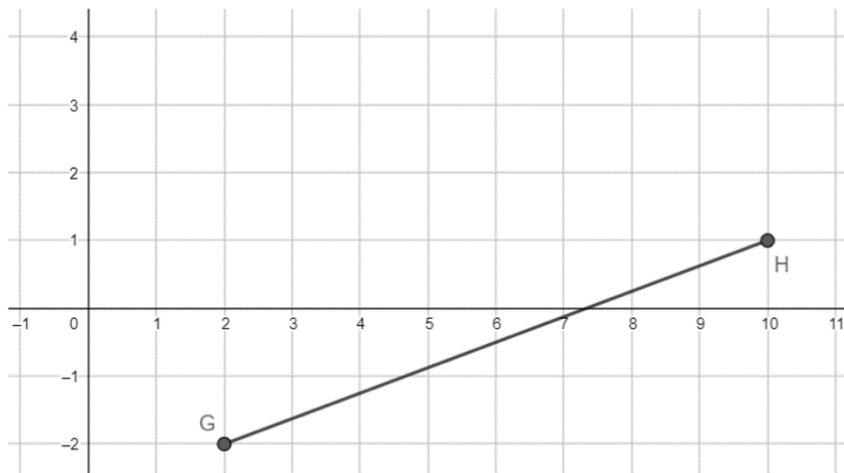
9. What is the vertical distance of $\overline{VR}$?

$$\frac{1}{4}(7 - 4) = \frac{3}{4}$$

10. What are the coordinates of point R ?

$$(-8 + 3, 7 - \frac{3}{4}) = (-5, 6\frac{1}{4})$$

Use the following information for questions 1-2. Point G has coordinates $(2, -2)$ and point H has coordinates $(10, 1)$.



1. Complete the table to partition $\overline{GH}$ in a 3:1 ratio.

	x -value	y -value
$G(2, -2)$	$\frac{1}{4}(2) = \frac{1}{2}$	$\frac{1}{4}(-2) = -\frac{1}{2}$
$H(10, 1)$	$\frac{3}{4}(10) = 7\frac{1}{2}$	$\frac{3}{4}(1) = \frac{3}{4}$
Total	8	$\frac{1}{4}$

2. What is the midpoint of $\overline{GH}$?

$$\left(\frac{2+10}{2}, \frac{-2+1}{2}\right) = \left(6, -\frac{1}{2}\right)$$

Use the following information for questions 3-4. Line segment BT has endpoints at $B(-6, -4)$ and $T(4, 6)$.

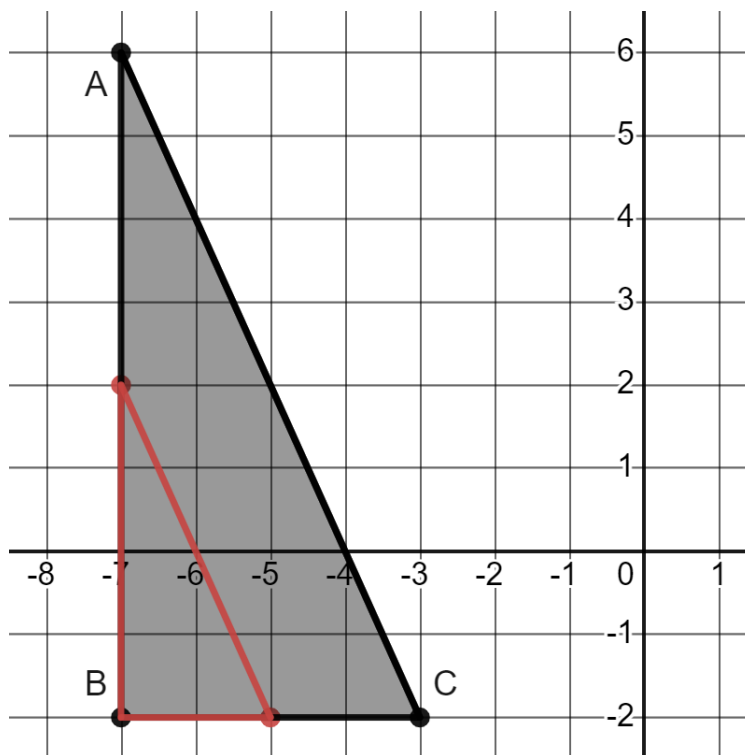
3. Write an expression for the point that partitions directed line segment BT in a 2:3 ratio.

$$\frac{3}{5}(\text{point } B) + \frac{2}{5}(\text{point } T)$$

4. Use the expression to find the point that partitions directed line segment BT in a 2:3 ratio.

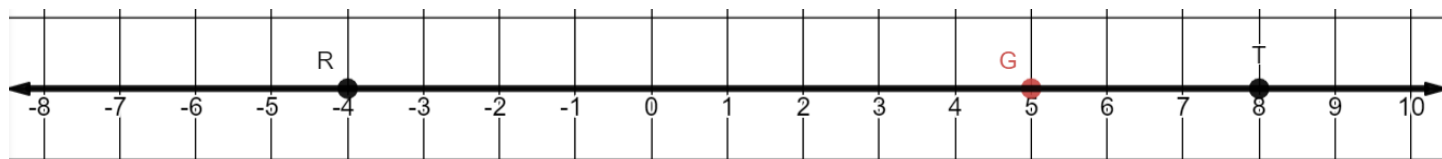
$$\left(\frac{3}{5}(-6) + \frac{2}{5}(4), \frac{3}{5}(-4) + \frac{2}{5}(6)\right) = (-2, 0)$$

Use the following information for question 5. Triangle ABC has vertices at $A(-7,6)$, $B(-7,-2)$, and $C(-3,-2)$.



5. Use segment partitioning to dilate triangle ABC using center B and scale factor $\frac{1}{2}$.

Use the following information for questions 6-7. A number line is shown. Each grid space is 1 unit.



6. Determine the point on the number line that satisfies the expression shown. Label the point G .

$$\frac{1}{4}(\text{point } R) + \frac{3}{4}(\text{point } T)$$

$$\frac{1}{4}(-4) + \frac{3}{4}(8) = 5$$

7. What is the ratio of $RG : GT$?

$$5 - (-4) : 8 - 5$$

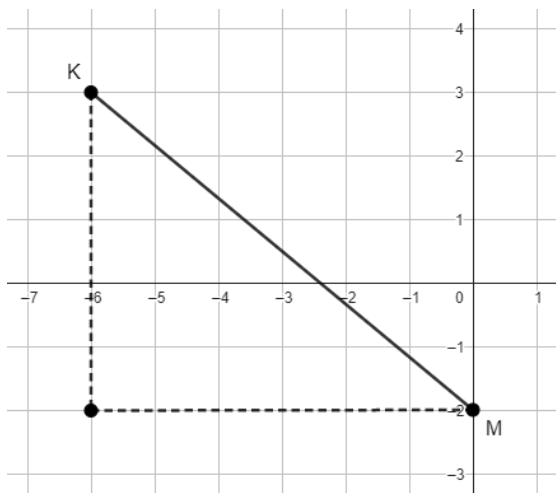
$$9 : 3$$

Geometry
Unit 13
Lesson 6 Homework

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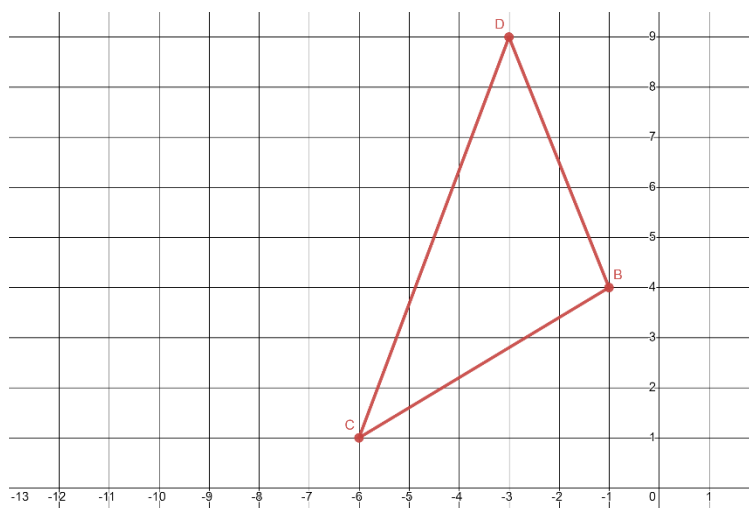
1. Find the length of $\overline{KM}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.

$$\sqrt{6^2 + 5^2} = \sqrt{61} \approx 7.810$$



Use the following information for questions 2-4. $\triangle BCD$ has vertices at $B(-1,4)$, $C(-6,1)$, and $D(-3,9)$.

2. Graph $\triangle BCD$ on the coordinate plane.



3. Find the length of each segment. If necessary, round to the nearest tenth.

Line Segment	Length of Line Segment
$\overline{BC}$	$\sqrt{3^2 + 5^2} = \sqrt{34} \approx 5.8$
$\overline{CD}$	$\sqrt{3^2 + 8^2} = \sqrt{73} \approx 8.5$
$\overline{BD}$	$\sqrt{2^2 + 5^2} = \sqrt{29} \approx 5.4$

4. Complete the statement.

$\triangle BCD$ is classified as a(n)

- ☐ equilateral
- ☐ isosceles
- ☒ scalene

triangle because $\triangle BCD$ has

- ☒ no
- ☐ two
- ☐ three

equal side lengths.

Use the following information for questions 5-6. $\triangle FGH$ has vertices at $F(-2,0)$, $G(2,5)$, $H(7,1)$.

5. Find the length of each segment. If necessary, round to the nearest tenth.

Line Segment	Length of Line Segment
$\overline{FG}$	$\sqrt{4^2 + 5^2} = \sqrt{41} \approx 6.4$
$\overline{GH}$	$\sqrt{5^2 + 4^2} = \sqrt{41} \approx 6.4$
$\overline{FH}$	$\sqrt{9^2 + 1^2} = \sqrt{82} \approx 9.1$

6. Complete the statement.

$\triangle FGH$ is classified as a(n)

- ☐ equilateral
- ☒ isosceles
- ☐ scalene

triangle because $\triangle FGH$ has

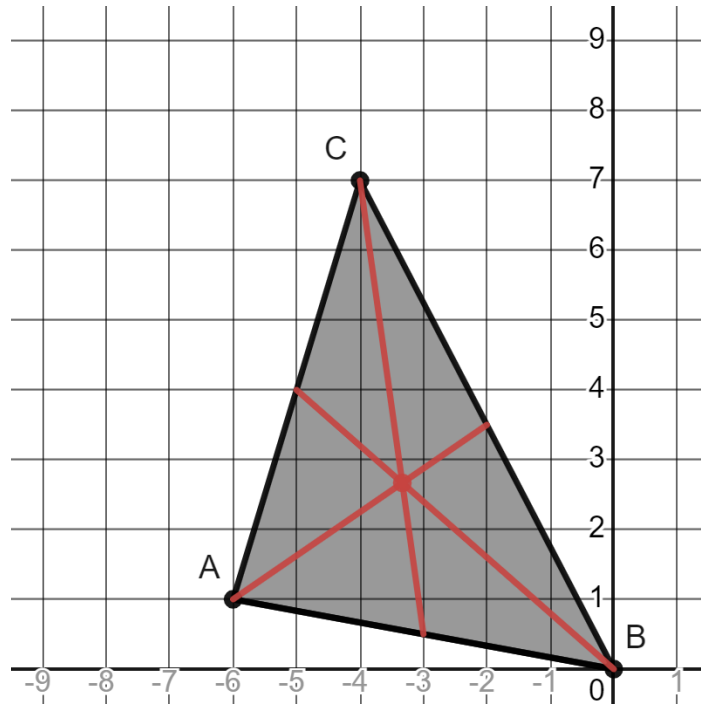
- ☐ no
- ☒ two
- ☐ three

equal side lengths.

Geometry
Unit 13
Lesson 7 Homework

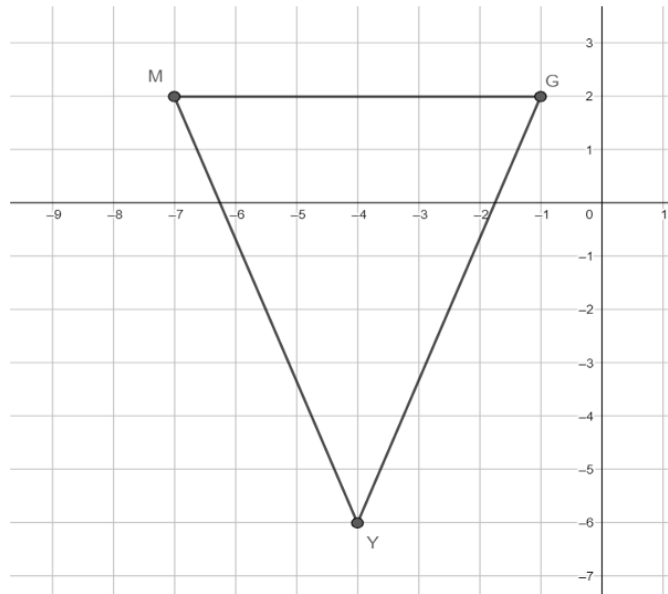
Name Answer Key
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Use the following information for questions 1-4. Triangle ABC has vertices at $A(-6,1)$, $B(0,0)$, and $C(-4,7)$.



1. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.
$$\frac{-6+0-4}{3} = -\frac{10}{3} \approx -3.333$$
2. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.
$$\frac{1+0+7}{3} = \frac{8}{3} \approx 2.667$$
3. What are the coordinates of the centroid?
 $(-3.333, 2.667)$
4. Verify the coordinates of the centroid by drawing the median of each side.
See graph.

Use the following information for question 5. Triangle GMY has vertices at $G(-1,2)$, $M(-7,2)$, and $Y(-4,-6)$.



5. What are the coordinates of the centroid of triangle GMY ?

$$\left(\frac{-1-7-4}{3}, \frac{2+2-6}{3}\right) = \left(-4, -\frac{2}{3}\right)$$

6. $\triangle KLM$ has vertices at $K(7,2)$, $L(6,9)$, and a centroid at $(5,6)$. What are the coordinates of vertex M ?

$$\frac{7+6+M_x}{3} = 5$$

$$13 + M_x = 15$$

$$M_x = 2$$

$$\frac{2+9+M_y}{3} = 6$$

$$11 + M_y = 18$$

$$M_y = 7$$

$(2,7)$

Keziah is finding the centroid of $\triangle ALG$ with vertices at $A(-1,3)$, $L(2,0)$, and $G(-4,-6)$.

Use the following information for questions 7-8. Her work is shown.

Keziah's Work	
$x = \frac{1+2+4}{3}$	$y = \frac{3+0+6}{3}$
$x = \frac{7}{3}$	$y = \frac{9}{3} = 3$
Centroid $\left(\frac{7}{3}, 3\right)$	

7. What is Keziah's mistake?

She changed all negative values to positive (A_x, G_x, G_y) before calculating the averages.

8. What are the coordinates of the centroid of $\triangle ALG$?

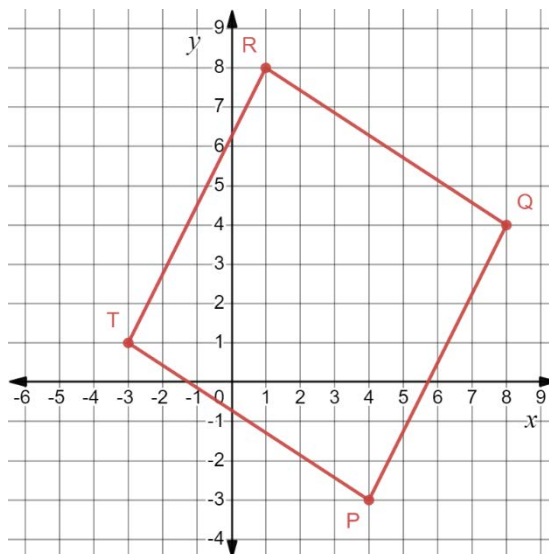
$$\left(\frac{-1+2-4}{3}, \frac{3+0-6}{3}\right) = (-1, -1)$$

Geometry
Unit 13
Lesson 8 Homework

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Use the following information for questions 1-4. $PQRT$ has vertices at $P(4, -3)$, $Q(8, 4)$, $R(1, 8)$, and $T(-3, 1)$.

1. Graph $PQRT$ on the coordinate plane.



2. Complete the table.

Line Segment	Length	Slope	Length of Diagonal Line Segment
$\overline{PQ}$	$\sqrt{7^2 + 4^2} = \sqrt{65}$	$\frac{4 - (-3)}{8 - 4} = \frac{7}{4}$	
$\overline{QR}$	$\sqrt{7^2 + 4^2} = \sqrt{65}$	$\frac{8 - 4}{1 - 8} = -\frac{4}{7}$	
$\overline{RT}$	$\sqrt{7^2 + 4^2} = \sqrt{65}$	$\frac{8 - 1}{1 - (-3)} = \frac{7}{4}$	
$\overline{PT}$	$\sqrt{7^2 + 4^2} = \sqrt{65}$	$\frac{1 - (-3)}{-3 - 4} = -\frac{4}{7}$	
$\overline{PR}$			$\sqrt{11^2 + 3^2} = \sqrt{130}$
$\overline{QT}$			$\sqrt{11^2 + 3^2} = \sqrt{130}$

3. Circle all classifications that apply to $PQRT$.

Isosceles Trapezoid

Kite

Parallelogram

Rectangle

Rhombus

Square

Trapezoid

4. Complete the statement.

Since $PQRT$ ☐ has congruent diagonals, $PQRT$ is classified as a(n)
☐ does not have
 square.

Use the following information for question 5. $PRKS$ has vertices at $P(-3,4)$, $R(1,0)$, $K(6,-1)$, and $S(2,3)$.

5. $PRKS$ is classified as a parallelogram because...

$$PS = \sqrt{5^2 + 1^2} = \sqrt{26}$$

$$SK = \sqrt{4^2 + 4^2} = \sqrt{32}$$

$$RK = \sqrt{5^2 + 1^2} = \sqrt{26}$$

$$PR = \sqrt{4^2 + 4^2} = \sqrt{32}$$

Slopes:

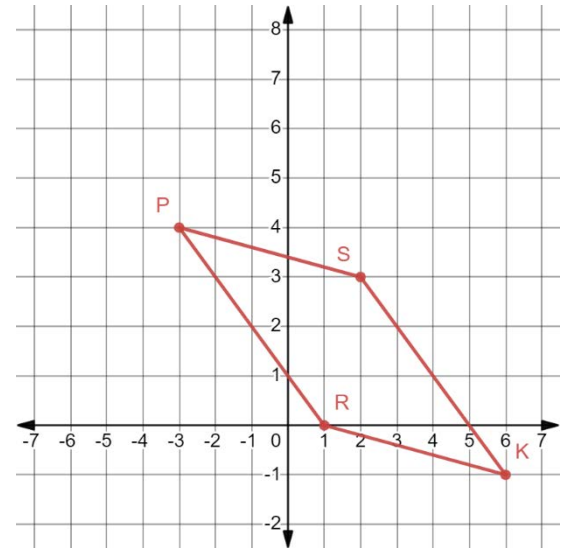
$$\overline{PS}: -\frac{1}{5}$$

$$\overline{SK}: -1$$

$$\overline{RK}: -\frac{1}{5}$$

$$\overline{PR}: -1$$

Opposite sides, but not all sides, congruent.
 Opposite sides parallel.



Use the following information for question 6-7. Saniya graphed $SMRT$ and classified the quadrilateral as a kite.

6. Do you agree with Saniya? No.

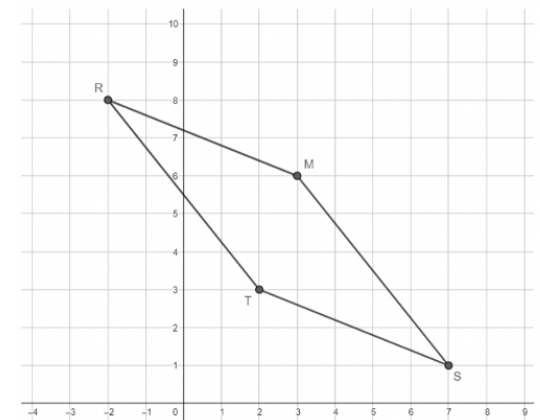
$$RM = \sqrt{5^2 + 2^2} = \sqrt{29}$$

$$ST = \sqrt{5^2 + 2^2} = \sqrt{29}$$

$$SM = \sqrt{5^2 + 4^2} = \sqrt{41}$$

$$RT = \sqrt{5^2 + 4^2} = \sqrt{41}$$

No adjacent sides are congruent, so not a kite.



7. Justify your answer using properties and showing your work.
 Side lengths show that opposite sides are congruent, but adjacent sides are not.

Geometry
Unit 13
Lesson 9 Homework

Name Answer Key
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1. Samal is using a computer software to design the layout of his new game room. The base of the new game room creates $GAMR$ with coordinates $G(-7, -5)$, $A(3, -9)$, $M(5, -4)$, and $R(-5, 0)$.

- a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{GA}$	$\sqrt{(3 - (-7))^2 + (-9 - (-5))^2} = \sqrt{10^2 + (-4)^2} = \sqrt{100 + 16}$ $= \sqrt{116} = 2\sqrt{29}$
$\overline{AM}$	$\sqrt{(-9 - (-4))^2 + (3 - 5)^2} = \sqrt{(-5)^2 + (-2)^2} = \sqrt{25 + 4} = \sqrt{29}$
$\overline{MR}$	$\sqrt{(0 - (-4))^2 + (-5 - 5)^2} = \sqrt{4^2 + (-10)^2} = \sqrt{16 + 100} = \sqrt{116}$ $= 2\sqrt{29}$
$\overline{GR}$	$\sqrt{(0 - (-5))^2 + (-5 - (-7))^2} = \sqrt{5^2 + 2^2} = \sqrt{25 + 4} = \sqrt{29}$

- b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{GM}$	$\sqrt{(-5 - (-4))^2 + (-7 - 5)^2} = \sqrt{(-1)^2 + (-12)^2} = \sqrt{1 + 144}$ $= \sqrt{145}$
$\overline{AR}$	$\sqrt{(-9 - 0)^2 + (3 - (-5))^2} = \sqrt{(-9)^2 + 8^2} = \sqrt{81 + 64} = \sqrt{145}$

- c. Finish the statement.

$GAMR$ is classified as a **rectangle** because ... **opposite sides are congruent and diagonals are congruent.**

- d. If 1 unit on the coordinate plane equals 4 feet (ft), what are the dimensions of the base of the new game room? Round to the nearest thousandth.

$$GA = 2\sqrt{29} \approx (10.7703)4 = 43.081 \text{ ft}$$

$$AM = \sqrt{29} \approx (5.38516)4 = 21.541 \text{ ft}$$

2. $STLE$ has vertices at $S(-5,6)$, $T(0,-3)$, $L(6,-5)$, and $E(1,4)$.

$STLE$ is classified as a(n) **parallelogram** because... **opposite sides congruent, but diagonals are not.**

3. Franco is designing a template for his new goat pen. He created two quadrilaterals using a computer software program.

$MFCR$	$AMBT$
$M(-1,5)$, $F(-9,-3)$, $C(-5,-7)$, $R(3,1)$	$A(-2,-3)$, $M(4,5)$, $B(0,7)$, $T(-6,0)$

- a. Which quadrilateral can be classified as a rectangle? Show your work.

$MFCR$

$$MR = \sqrt{(3 - (-1))^2 + (1 - 5)^2} = \sqrt{4^2 + 4^2} = \sqrt{16 + 16} = 4\sqrt{2}, \text{ slope} = -1$$

$$FC = \sqrt{(-7 - (-3))^2 + (-5 - (-9))^2} = \sqrt{4^2 + 4^2} = \sqrt{16 + 16} = 4\sqrt{2}, \text{ slope} = -1$$

$$CR = \sqrt{(-7 - 1)^2 + (-5 - 3)^2} = \sqrt{8^2 + 8^2} = \sqrt{64 + 64} = 8\sqrt{2}, \text{ slope} = 1$$

- b. List the properties of a rectangle that you used.

Opposite sides congruent and parallel, $\overline{MR} \parallel \overline{FC}$, $\overline{MR} \cong \overline{FC}$, adjacent sides perpendicular by comparing slope.

- c. If 1 unit of the coordinate plane equals 7 feet (ft), what are the measurements of the sides of the rectangle?

$$7(4\sqrt{2}) = 28\sqrt{2} \approx 39.6 \text{ ft}$$

$$7(8\sqrt{2}) = 56\sqrt{2} \approx 79.2 \text{ ft}$$

4. Akilah was graphing square $STVE$, but mixed up her vertices. She knows that point S has coordinates $(0,4)$ and point E has coordinates $(8,0)$. Her two options for points T and V are shown.

Point T	Point V
$(-4, -4)$	$(-2, -8)$
$(4, -4)$	$(4, -8)$

- a. Which coordinates are correct for points T and V ? Justify your answer or show your work.

$$SE = 4\sqrt{5}, \text{ slope} = -\frac{1}{2}$$

$$ST = 4\sqrt{5}, \text{ slope} = 2$$

$$EV = 4\sqrt{5}, \text{ slope} = 2$$

$$T(-4, -4), V(4, -8)$$

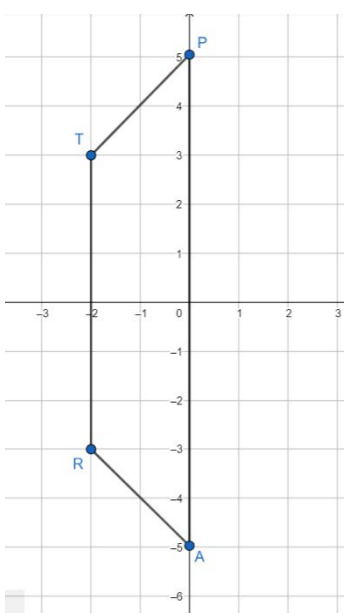
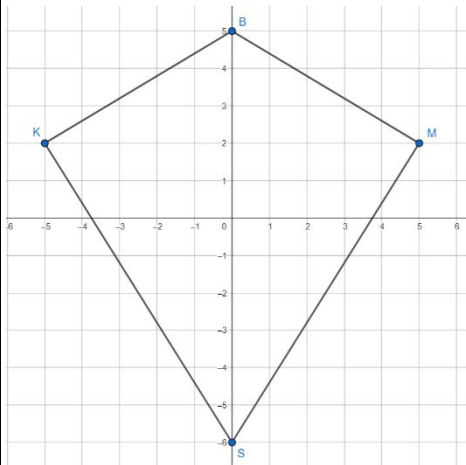
- b. Which properties of a square did you use to determine the correct coordinates for points T and V ?

Length of all 4 sides are congruent and slope of adjacent sides opposite reciprocal (sides perpendicular).

Geometry
Unit 13
Lesson 10 Homework

Name Answer Key
Date _____
Period _____

Complete the table for each given image.

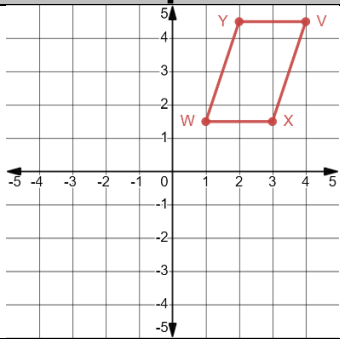
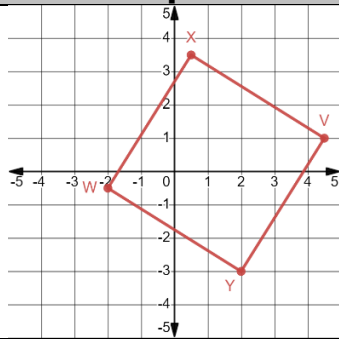
Question 1	Work		Answer
			Properties
	Slope	$\overline{TP}: 1, \overline{RA}: -1$ $\overline{PA}: \text{undefined}$ $\overline{TR}: \text{undefined}$	Congruent legs. Congruent diagonals. One set of parallel sides.
	Side Lengths	$RT = 6$ $TP = \sqrt{2^2 + 2^2} = 2\sqrt{2}$ $PA = 10$ $AR = \sqrt{2^2 + 2^2} = 2\sqrt{2}$	
	Diagonal Lengths	$RP = \sqrt{2^2 + 8^2} = 2\sqrt{17}$ $TA = \sqrt{2^2 + 8^2} = 2\sqrt{17}$	Quadrilateral Isosceles trapezoid
Question 2	Work		Answer
			Properties
	Slope	$\overline{KB}: \frac{3}{5}, \overline{MS}: \frac{8}{5}$ $\overline{BM}: -\frac{3}{5}, \overline{KS}: -\frac{8}{5}$	Two sets of adjacent congruent sides. No parallel sides.
	Side Lengths	$KB = \sqrt{3^2 + 5^2} = \sqrt{34}$ $BM = \sqrt{3^2 + 5^2} = \sqrt{34}$ $MS = \sqrt{8^2 + 5^2} = \sqrt{89}$ $KS = \sqrt{8^2 + 5^2} = \sqrt{89}$	
	Diagonal Lengths	$SB = 11$ $KM = 10$	Quadrilateral Kite

For questions 3 and 4, given the coordinates, classify the quadrilaterals based on their properties.

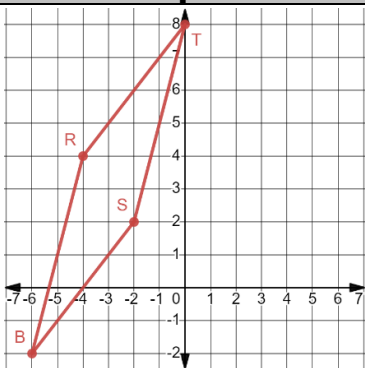
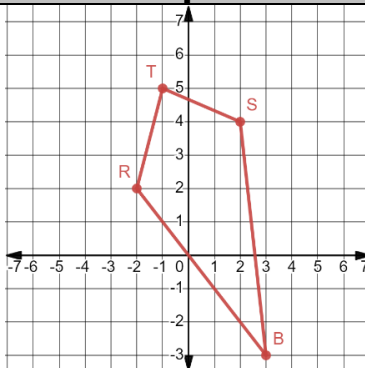
Question 3	Work		Answer
			Properties
$SKWC$ has vertices at $S(6, -3)$, $K(1, 3)$, $W(-5, -2)$, and $C(0, -8)$. $SKWC$ is classified as a(n) square because... 4 congruent sides and 4 right angles	Slope	$\overline{SK}: -\frac{6}{5}, \overline{WC}: -\frac{6}{5}$ $\overline{KW}: \frac{5}{6}, \overline{CS}: \frac{5}{6}$	Perpendicular consecutive sides (right angles). 4 congruent sides and congruent diagonals. 2 sets of parallel sides.
	Side Lengths	$SK = \sqrt{5^2 + 6^2} = \sqrt{61}$ $KW = \sqrt{5^2 + 6^2} = \sqrt{61}$ $WC = \sqrt{5^2 + 6^2} = \sqrt{61}$ $CS = \sqrt{5^2 + 6^2} = \sqrt{61}$	
	Diagonal Lengths	$SW = \sqrt{11^2 + 1^2} = \sqrt{122}$ $KC = \sqrt{11^2 + 1^2} = \sqrt{122}$	Quadrilateral Square
Question 4	Work		Answer
			Properties
$GNTH$ has vertices at $G(5, 8)$, $N(1, 1)$, $T(0, -7)$, and $H(4, 0)$. $GNTH$ is classified as a(n) rhombus because... 4 congruent sides with no right angles.	Slope	$\overline{GN}: \frac{7}{4}, \overline{TH}: \frac{7}{4}$ $\overline{NT}: \frac{8}{1}, \overline{HG}: \frac{8}{1}$	Two sets of parallel opposite sides. All 4 sides congruent. No perpendicular sides. Diagonals not congruent.
	Side Lengths	$GN = \sqrt{4^2 + 7^2} = \sqrt{65}$ $NT = \sqrt{1^2 + 8^2} = \sqrt{65}$ $TH = \sqrt{4^2 + 7^2} = \sqrt{65}$ $HG = \sqrt{1^2 + 8^2} = \sqrt{65}$	
	Diagonal Lengths	$GT = \sqrt{5^2 + 15^2} = 5\sqrt{10}$ $NH = \sqrt{3^2 + 1^2} = \sqrt{10}$	Quadrilateral Rhombus

Plot the given coordinates in to order to compare each pair of graphs. Record your work and show your justifications in the table.

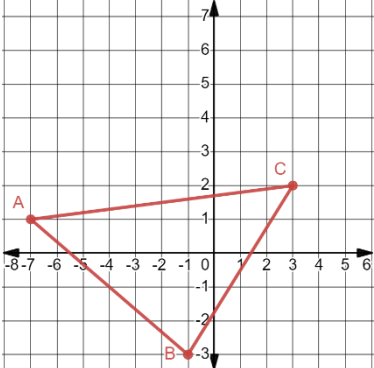
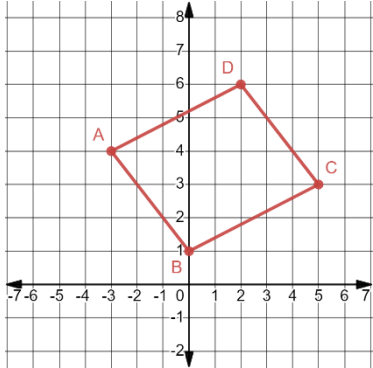
1.

Graph 1	Graph 2
	
Coordinates: $V(4,4.5), X(3,1.5), W(1,1.5), Y(2,4.5)$	Coordinates: $V(4.5,1), X(0.5,3.5), W(-2, -0.5), Y(2, -3)$
Classification: Parallelogram	Classification: Square
Property: Opposite sides are parallel and congruent. Diagonals bisect each other.	Property: All sides congruent. Opposite sides parallel. Diagonals congruent & bisect each other.
Justification: Slopes of $\overline{WY}$ & $\overline{XZ}$ are zero. Slopes of $\overline{WX}$ & $\overline{YZ}$ are 3. $WX = YZ = 2$ $WY = XZ = \sqrt{3^2 + 1^2} = \sqrt{10}$	Justification: $VX = XW = WY = YV = \sqrt{\left(\frac{5}{2}\right)^2 + 4^2} = \frac{\sqrt{89}}{2}$ Slopes $WX = YZ = \frac{8}{5} \perp$ to slopes $WY = XZ = -\frac{5}{8}$

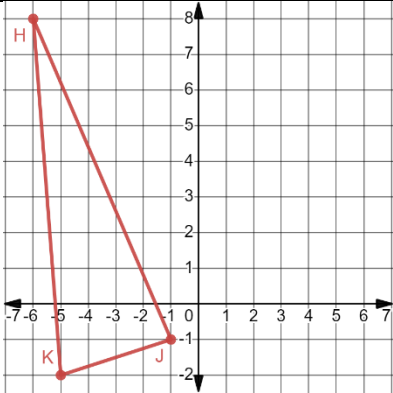
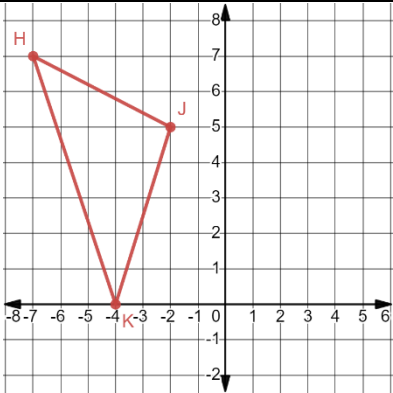
2.

Graph 1	Graph 2
	
Coordinates: $T(0,8), R(-4,4), B(-6, -2), S(-2,2)$	Coordinates: $T(-1,5), R(-2,2), B(3, -3), S(2,4)$
Classification: Parallelogram	Classification: Kite
Property: Opposite sides are parallel and congruent. Diagonals bisect each other.	Property: Perpendicular diagonals. Two pairs of congruent adjacent sides.
Justification: $BR = ST = 2\sqrt{10}, TR = SB = 4\sqrt{2}$ Slopes $BR = ST = 3$ Slopes $TR = SB = 1$	Justification: Slope $TB = -2$; Slope $RS = \frac{1}{2}$ $RB = SB = 5\sqrt{2}$ $RT = TS = \sqrt{10}$

3.

Graph 1	Graph 2
	
Coordinates: $A(-7, 1)$, $B(-1, -3)$, $C(3, 2)$	Coordinates: $A(-3, 4)$, $B(0, 1)$, $C(5, 3)$, $D(2, 6)$
Classification: Scalene triangle	Classification: Parallelogram
Property: No sides congruent.	Property: Opposite sides are parallel and congruent.
Justification: $AB = 2\sqrt{13}$ $BC = \sqrt{41}$ $AC = \sqrt{101}$	Justification: $Slopes\ AD = BC = \frac{2}{5}$ $Slopes\ AB = DC = -1$ $AB = DC = 3\sqrt{2}$; $AD = BC = \sqrt{29}$

4.

Graph 1	Graph 2
	
Coordinates: $H(-6, 8)$, $J(-1, -1)$, $K(-5, -2)$	Coordinates: $H(-7, 7)$, $J(-2, 5)$, $K(-4, 0)$
Classification: Scalene triangle	Classification: Isosceles right triangle
Property: No sides congruent.	Property: Exactly two sides are congruent. Two sides are perpendicular.
Justification: $HJ = \sqrt{5^2 + 9^2} = \sqrt{106}$ $JK = \sqrt{1^2 + 4^2} = \sqrt{17}$ $HK = \sqrt{1^2 + 10^2} = \sqrt{101}$	Justification: $HJ = \sqrt{2^2 + 5^2} = \sqrt{29}$ $JK = \sqrt{2^2 + 5^2} = \sqrt{29}$ $HK = \sqrt{3^2 + 7^2} = \sqrt{58}$ $Slope\ HJ = -\frac{2}{5}$; $Slope\ KJ = \frac{5}{2}$

1. Right triangle PYZ has vertices at $P(9,7)$, $Y(3,-1)$, $Z(9,-1)$. Right triangle RST has vertices at $R(6,-2)$, $S(1,4)$, $T(1,-2)$.

- a. Determine if $\triangle PYZ$ and $\triangle RST$ are congruent, similar, or neither.

$$PY = 10$$

$$YZ = 6$$

$$PZ = 8$$

$$RS = \sqrt{61}$$

$$ST = 6$$

$$TR = 5$$

neither

- b. What definition, theorem, or property justifies your conclusions?

Answers may vary. One possible answer: Since none of the sides in $\triangle RST$ are congruent to PY nor PZ , the shapes are not congruent based on definition of congruence. Since the pairs of corresponding sides are not proportional, the triangles are not similar.

- c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.

Using RST as the preimage, reflect $\triangle RST$ on $\overline{ST}$, then translate 8 to the right and 1 up, $(x,y) \rightarrow (x+7,y+1)$ to place T onto . Mapping shows $\triangle RST$ and $\triangle PYZ$ are neither congruent nor similar.

2. Determine if the given polygons are congruent, similar, or neither.

- a. Trapezoid $FDGK$: $F(-8,-2)$, $D(-4,-2)$, $G(-2,-8)$, $K(-10,-6)$.
Trapezoid $MPRT$: $M(8,2)$, $P(4,2)$, $R(2,8)$, $K(10,6)$

☒ Congruent

☐ Similar

☐ Neither

- b. Kite $ABCD$: $A(-1,1)$, $B(4,1)$, $C(5,4)$, $D(2,5)$.
Kite $KFRP$: $K(0,0)$, $F(-1,-4)$, $R(-3,-3)$, $P(-4,-1)$.

- ☐ Congruent
☐ Similar
☒ Neither

- c. Rectangle $JKMP$: $J(-4,4)$, $K(0,-2)$, $M(3,0)$, $P(-1,6)$.
Rectangle $BCDF$: $B(8,6)$, $C(0,8)$, $D(-1,4)$, $F(7,2)$

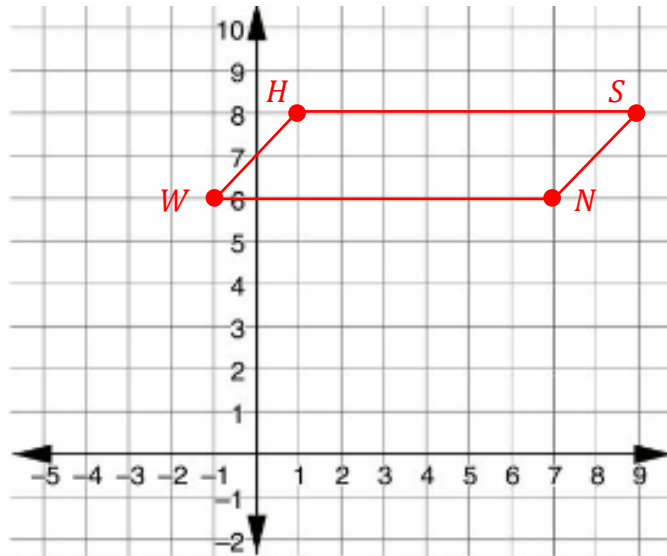
- ☐ Congruent
☒ Similar
☐ Neither

3. Hannah and Zuki are determining if parallelograms $NPSE$ and $FGHR$ are congruent, similar, or neither. Parallelogram has vertices at $N(-9,2)$, $P(-1,5)$, $S(1,9)$, and $E(-7,6)$. Parallelogram $F(-4.5,1)$, $G(-0.5,2.5)$, $H(0.5,4.5)$, and $R(-3.5,3)$. Their work is shown.

Hannah	Zuki
Parallelograms $NPSE$ and $FGHR$ are similar because $\frac{NE}{FR} = \frac{PS}{GH}$	Parallelograms $NPSE$ and $FGHR$ are not similar because $\frac{NP}{FG} = \frac{SE}{HR}$

- a. Who do you agree with?
Hannah
- b. Explain the mistake that was made and how to correct it.
Zuki stated they are not similar and they are.

1. Parallelogram $SHWN$ has vertices at $S(9,8)$, $H(1,8)$, $W(-1,6)$, and $N(7,6)$.



- a. Graph $SHWN$.
- b. Complete the table.

Line Segment	Length
$\overline{SH}$	8
$\overline{HW}$	$\sqrt{2^2 + 2^2} = 2\sqrt{2}$
$\overline{WN}$	8
$\overline{SN}$	$\sqrt{2^2 + 2^2} = 2\sqrt{2}$

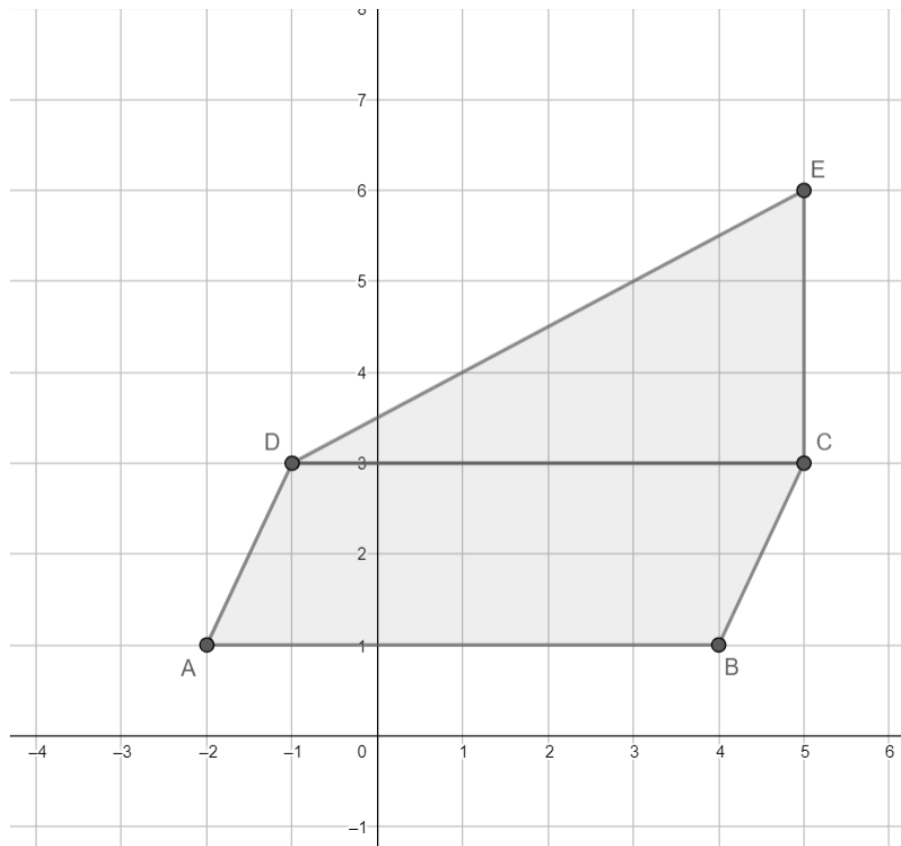
- c. What is the perimeter of $SHWN$?

$$2(2\sqrt{2}) + 2(8) = 4\sqrt{2} + 16 \approx 21.66$$

- d. What is the area of $SHWN$?

$$2 \cdot 8 = 16$$

2. The composite figure shown is created by $A(-2,1)$, $B(4,1)$, $C(5,3)$, $D(-1,3)$, and $E(5,6)$.



- a. Complete the table. Round your answer to the nearest thousandth.

Figure	Perimeter	Area
$\triangle CDE$	$6 + 3 + \sqrt{6^2 + 3^2} \approx 15.708$	$\frac{1}{2}(3)(6) = 9$
$ABCD$	$2\sqrt{1^2 + 2^2} + 2(6) \approx 16.472$	$2 \cdot 6 = 12$

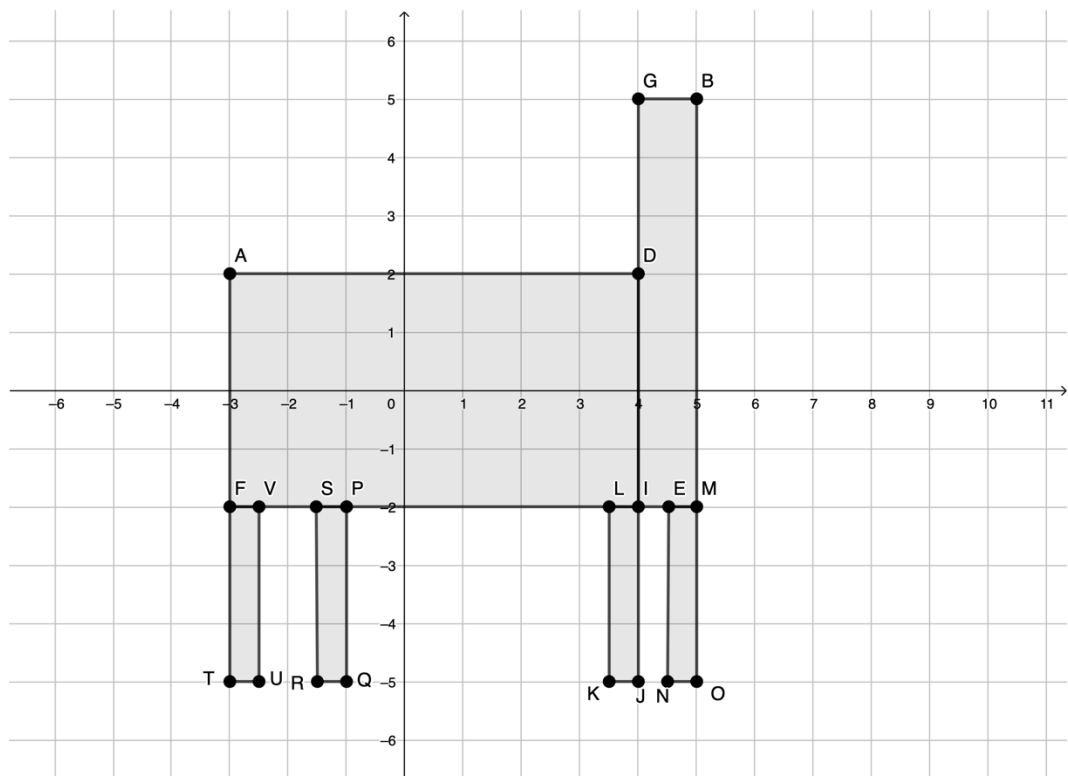
- b. What is the perimeter of the composite figure?

$$\sqrt{5} + \sqrt{45} + 3 + \sqrt{5} + 6 \approx 20.180$$

- c. What is the area of the composite figure?

$$9 + 12 = 21$$

1. Brandon is creating a layout for his new arcade game. Each part of the model is made of quadrilaterals. The front legs, $LIJK$ & $EMON$, and the back legs, $FVUT$ & $SPQR$, are all congruent rectangles. $GBMI$ and $ADIF$ are rectangles.

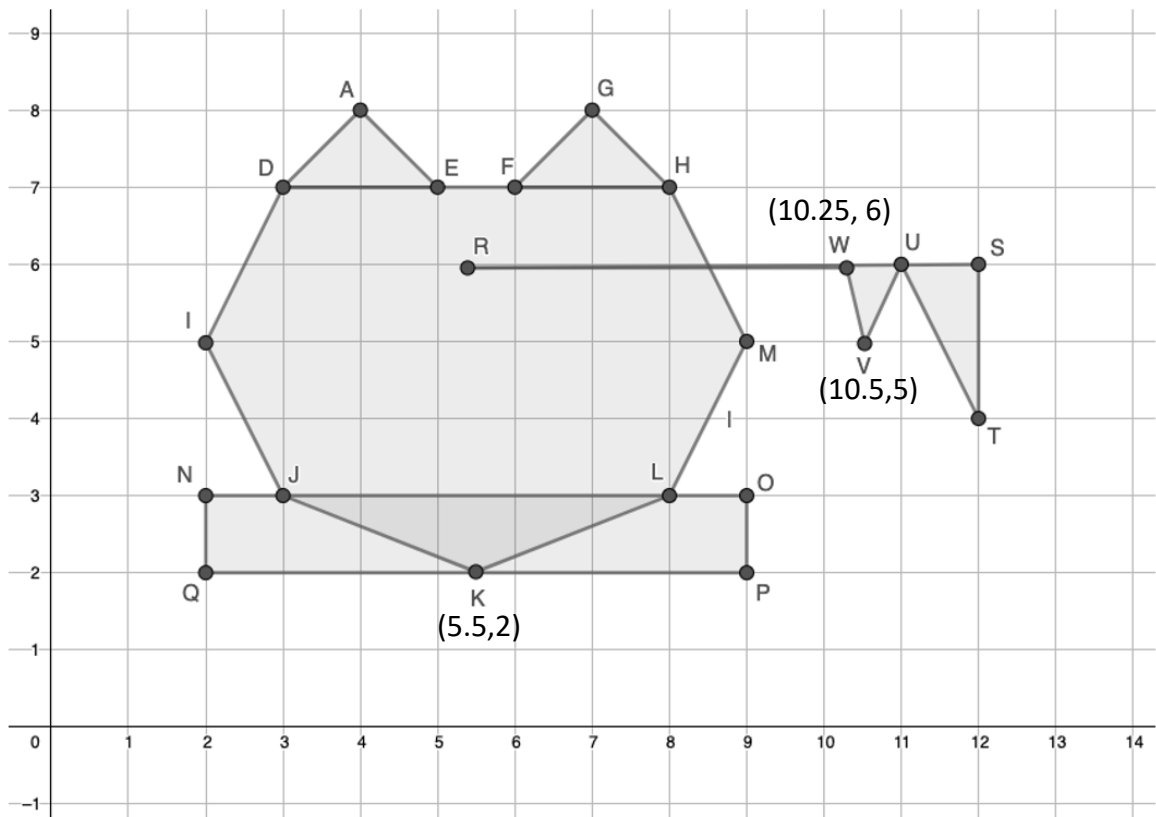


- a. Complete the table. If necessary, round to the nearest thousandth.

Section	Perimeter	Area
Front legs $LIJK, EMON$	7 each, 14 total	1.5 each, 3 total
Back legs $FVUT, SPQR$	7 each, 14 total	1.5 each, 3 total
Rectangle $GBMI$	16	7
Rectangle $ADIF$	22	28

- b. What is the perimeter of the composite figure made by all of the sections?
If necessary, round to the nearest thousandth.
 $66 - 8(0.5) - 2(4) = 54$ units
- c. What is the area of the composite figure made by all of the sections?
If necessary, round to the nearest thousandth.
41 square units
- d. If each square unit weights 20 lbs. How much will the arcade game weigh?
 $41(20) = 820$ lbs.

2. Asia is building her very first Battle Bot. She makes her first sketch of her battle bot's profile on the grid. Each square on the grid represents 1 inch.



- a. Complete the table. If necessary, round to the nearest thousandth.

Section	Area (in ²)	Perimeter (in)
Tail cover <i>NQPO</i>	7	16
Triangles <i>ADE</i> and <i>GFH</i>	1 each, 2 total	≈ 4.828 each, ≈ 9.567 total
Triangle <i>STU</i>	1	≈ 5.236
Triangle <i>WUV</i>	0.375	0.75 + 1.031 + 1.118 = 2.899
Battle Bot Body <i>DIJKLMH</i>	26.5	≈ 19.329

- b. If every unit square is equal to 2.5 pounds, what is the total weight of the Battle Bot?
36.875(2.5) ≈ 92.188 lbs.